

0.5Li₂SO₄-ZrCl₄ — experiment-structure-MLIP comparison

A sulfate-containing glass tests transport and local structure simultaneously

1 Literature

Tang et al. (2026)
Experiment + scattering
AIMD and tuned MACE

2 Structure

Cluster-based packing
Li₃₂Zr₃₂Cl₁₂₈S₁₆O₆₄
272 atoms; NEP89

3 MD protocol

320 / 330 / 340 / 350 K
50 ps NVT equilibration
300 ps NVT production

4 Analysis

MSD, D(T), σ NE and E_a
RDF and coordination
Li-O environment / mobility

5 Conclusion

Sulfate motif retained
Transport fit is literature-close
Long-range diffusion limited